

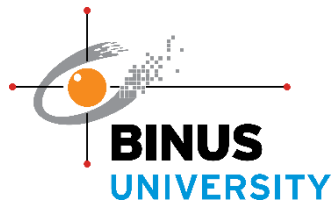
INDEPENDENT RESEARCH PROPOSAL

LIGHTWEIGHT TRANSFORMER AND CNN MODELS COMPARISON ANALYSIS FOR DEGENERATIVE LUMBAR SPINE CLASSIFICATION

Topic: Medical Image Classification

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CHAPTER 1

INTRODUCTION

1.1 Background

Low back pain is the leading cause of years lived with disability (YLDs) worldwide and is becoming a significant issue in an aging population. The primary contributors to mechanical lumbar spine disorders for which are responsible for up to 97% of low back pain diagnoses include degenerative intervertebral disc issues, lumbar spondylotic radiculopathy, spondylolisthesis, and spinal stenosis, all of which are part of a degenerative process affecting the lumbar spine (Park et al., 2021, p. 1).

A significant challenge in managing low back pain is that approximately 85 percent of patients with isolated low back pain do not receive a precise pathoanatomical diagnosis (Marcellino et al., 2021, p. 37). This lack of clarity complicates treatment decisions, as the relationship between symptoms reported by patients and the findings from imaging studies is often weak. Many individuals may present with similar symptoms, yet their imaging results can reveal diverse and sometimes ambiguous findings (Magayane, 2021).

Therefore, there will be a need for an accurate classification of the degenerative lumbar spine condition. It is essential to ensure timely diagnosis and implement effective treatment strategies. By correctly identifying the severity of the condition, healthcare providers can prioritize which patients require more careful evaluation and intervention, enhancing patient outcomes. This targeted approach ensures that those with more severe conditions receive timely and appropriate care, while also optimizing resource allocation within the healthcare system.

Alongside the rapid development of technology, artificial intelligence (AI), particularly deep learning, has become an increasingly vital tool in healthcare, especially for medical image processing which include computer vision, pattern recognition, image mining, and machine learning in several directions. Deep learning techniques, especially convolutional neural networks (CNNs), have demonstrated exceptional capabilities in analyzing and interpreting complex medical images such as MRI and CT scans. These models can automatically extract relevant features and patterns from imaging data (Suganyadevi et al., 2022, p. 19).

Thus, many computer-aided detection and diagnosis (CAD) schemes have been developed and tested to help clinicians read medical images more efficiently and make more accurate and objective diagnostic decisions (Chen et al., 2022, p.102444). One of them can be found in the research conducted by Samir S. Yadav and Shivajirao M. Jadhav in 2019. The study titled “Deep Convolutional Neural Network-Based Medical Image Classification for Disease Diagnosis”

provides a comparative analysis of VGG16 and InceptionV3 models for classifying pneumonia using chest X-ray images.

Moreover, the emergence of transformer-based models has significantly transformed the field of image analysis. Initially developed for natural language processing (NLP), these models have demonstrated their effectiveness as a compelling alternative to CNNs in computer vision, achieving performance levels comparable to those of CNNs (Gai et al., 2024, p. 59253). Transformers leverage a powerful mechanism called attention, which allows them to focus on specific parts of the input for enhanced results. This approach has led to the exploration of various strategies for applying transformers to images, including splitting images into patches and treating them as tokens. Which has resulted in competitive performance on tasks like image classification (Bazi et al., 2021, p. 516) such as in 2023, Ren et al. proposed using Transformer for COVID19 medical image classification. The proposed model was a hybrid of ResNet and ViT, named RMT-Net which resulted in a validation accuracy of 98.84% on the X-Ray dataset and 99.24% on the CT scans dataset.

1.2 Problem Formulation

This research aims to address the following key questions:

- a) How do lightweight transformer models perform in comparison to traditional CNN models in the classification of degenerative lumbar spine conditions?
- b) How do the model sizes and training times compare between lightweight transformers and CNNs in this classification task?

1.3 Scope

This research aims to enhance the performance of various models in the classification of degenerative lumbar spine conditions using magnetic resonance imaging (MRI) data. Specifically, the study focuses on comparing the effectiveness of CNN-based models, including Inception-V4, EfficientNet, and EfficientNetV2, against a lightweight transformer model utilizing the Vision Transformer (ViT) architecture. The primary objective is to evaluate the models' ability to classify degenerative spine conditions into three categories: normal, moderate, and severe. By analysing their performance, this research seeks to determine which approach offers superior accuracy and robustness in diagnosing degenerative lumbar spine conditions, thereby contributing to the growing body of knowledge in medical imaging and machine learning applications in healthcare.

The dataset utilized in this study was curated and annotated by the Radiological Society of North America (RSNA) and the American Society of Neuroradiology (ASNR). It includes MRI scans with severity scores (Normal/Mild, Moderate, or Severe) for five lumbar spine conditions across intervertebral disc levels L1/L2 to L5/S1 such as, Left Neural Foraminal Narrowing, Right Neural Foraminal Narrowing, Left Subarticular Stenosis, Right Subarticular Stenosis, and Spinal Canal Stenosis. The dataset was sourced from multiple international institutions, providing a diverse and representative sample of degenerative spine conditions.

To ensure adherence to research ethics, the dataset utilized in this study is obtained from the official Kaggle platform, specifically titled “RSNA 2024 Lumbar Spine Degenerative Classification”. The dataset, curated by experts from multiple international institutions, provides a robust foundation for this research. The methodology for evaluating the performance of the modified Inception-V4 and EfficientNet models will focus on accuracy as the primary metric.

1.4 Objectives and Benefits

1.4.1 Objectives

- a) To compare the performance of lightweight transformer models and traditional CNN models in the classification of degenerative lumbar spine conditions using magnetic resonance imaging (MRI) data.
- b) To evaluate and contrast the model sizes and training times between lightweight transformer models and CNNs in the context of classifying degenerative lumbar spine conditions.

1.4.2 Benefits

From these objectives, the potential benefits of conducting this research are as follows:

- a) Contributing to the advancement of knowledge in the field of machine learning applied to healthcare, thereby fostering the development of insights and strategies that can benefit both researchers and healthcare professionals. The performance improvements can assist healthcare professionals by providing a reliable tool for classifying lumbar spine health, potentially reducing diagnostic errors, and improving patient outcomes. This, in turn, leads to better-targeted treatment plans and improved patient outcomes, highlighting the model’s potential impact in delivering high-quality, effective healthcare.

- b) By comparing the performance of lightweight transformer models and traditional CNN models, this research aims to highlight the differences in classification accuracy and computational efficiency for degenerative lumbar spine conditions. The findings will provide insights into the scalability of these models for real-time diagnostics. Additionally, this study contributes to the academic community's understanding of the strengths and limitations of each model type, fostering the development of new strategies and insights that can benefit researchers, practitioners, and healthcare systems.

1.5 Methods

This study follows a systematic approach for the image classification task, starting with data collection, where MRI images relevant to degenerative lumbar spine conditions are gathered from the Kaggle dataset titled "RSNA 2024 Lumbar Spine Degenerative Classification." The dataset, curated and annotated by the Radiological Society of North America (RSNA) and the American Society of Neuroradiology (ASNR), includes MRI scans labelled with severity levels (Normal/Mild, Moderate, or Severe) for various degenerative conditions affecting intervertebral discs from L1/L2 to L5/S1. Following data collection, exploratory data analysis (EDA) is performed to evaluate class distributions, assess image quality, and identify initial patterns that inform the subsequent preprocessing steps. Key preprocessing tasks include resizing images for uniform dimensions, converting them to grayscale, transforming the data into tensor format for compatibility with the models, and normalizing pixel values to ensure consistency across the dataset.

Once the preprocessing is complete, the data is divided into training, validation, and testing sets to facilitate effective model development. During the modeling phase, both CNN-based models including Inception-V4, EfficientNet, and EfficientNetV2 and a transformer-based model using the Vision Transformer (ViT) architecture are trained. These models are then evaluated and compared based on their classification performance for degenerative lumbar spine conditions. The evaluation process primarily utilizes accuracy as the key performance metric, supported by calculations of True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN) values. This multifaceted evaluation framework allows for a comprehensive understanding of each model's effectiveness, guiding the selection of the most suitable architecture for this classification task.

1.6 Writing System

1. Title
2. Table of Contents
3. Chapter I – Introduction
 - A. Introduction – A big picture and background about the research topic.
 - B. Problem Formulation – Problem that will be investigate.
 - C. Scope – Research limitations and boundaries.
 - D. Objectives and Benefits – The purpose of the research and what benefits can be taken from the research.
 - E. Methods – Methods used in the research
 - F. Writing Systematics – the sequence or structure of writing scientific papers
4. Chapter 2 – Literature Review
 - A. Related Works – Comparison of this research with other similar studies.
 - B. Literature Review – Explanation of theories relevant to the research topic.
5. Chapter 3 - Methods
 - A. Thinking Framework – The steps that will be taken to solve the problem.
 - B. Research Methods – Detailed methods that will be used in the research.

CHAPTER 2

LITERATURE REVIEW

2.1 Related Works

a. Previous Work in Medical Image Classification Using CNN Based Model

A study conducted in 2019 by Samir S. Yadav et al. examined the performance of VGG16 and InceptionV3 architectures for pneumonia classification using chest X-ray images. Their dataset consists of three categories which are bacterial pneumonia, viral pneumonia, and normal cases. The models were primarily evaluated based on accuracy as the performance metric where the results indicated that VGG16 consistently outperformed InceptionV3 in terms of classification accuracy when identifying types of pneumonia, achieving an accuracy of 0.938 compared to InceptionV3's 0.869.

In 2024, Reddy et al. explored deep learning models for brain tumor detection and classification. The research compared several CNN architectures, including ResNet-50, VGG16, Inception V3, and a custom CNN model. They were using a dataset of 3,265 MRI images covering gliomas, meningiomas, pituitary tumors, and tumor-free images. The models were primarily evaluated based on metrics such as accuracy, recall, loss, and AUC. Results indicated that the custom CNN model achieved superior performance, with a validation accuracy of 94.4%, an AUC of 97.53%, and a recall rate of 92.2% which outperforms the other architectures.

Agarwal et al. (2023) developed an Alzheimer's disease (AD) diagnostic model using the EfficientNet-B0 CNN with a fusion of end-to-end and transfer learning approach. The study used imaging data from the IXI and Alzheimer's Disease Neuroimaging Initiative (ADNI) databases, consisting of 245 T1W MRI scans of cognitively normal (CN) subjects, 229 scans of AD subjects, and 229 scans of stable mild cognitive impairment (sMCI) subjects. The EfficientNet-B0 model was implemented with 5-fold stratified cross-validation that achieved an average training accuracy of 95.29% across all folds.

In a 2022 study by Muhammad Saquib Nazir et al. which focused on breast cancer detection in mammogram images introduced a powerful hybrid model by combining a Convolutional Neural Network (CNN) with the Inception-V4 architecture CNN-Inception-V4. This model achieved 99.2% accuracy, 99.8% sensitivity, 96.3% specificity, and 97% F1-score that outperforms models like CNN, Inception-V4, and

ResNet-50. However, the model's design increases computational load which is a challenge for deployment on resource-constrained devices like clinical edge devices.

Marques et al. (2020) introduced a CNN-based medical decision support system using the EfficientNet architecture for COVID-19 detection in CT and X-ray images. EfficientNet outperformed models such as DarkCovidNET, VGG19, MobileNet v2, CapsNet, and nCOVnet, achieving an accuracy of 99.62% in binary classification (COVID-19 vs. normal) and 96.70% in multiclass classification (COVID-19, pneumonia, and normal). The system demonstrated high precision and recall, with an F1-score of 99.62% for binary and 97.11% for multiclass classification.

Evwiekpaefe and Amrevuawho (2023) used the EfficientNetV2 model to classify dermatologic manifestations of cardiovascular diseases into seven distinct categories. The model was trained on a dataset of 2,665 images, including categories like cyanosis, liverdo reticularis, and xanthoma. Data augmentation was applied to expand training samples. By using transfer learning, EfficientNetV2 achieved 96.04% accuracy. While it excelled in capturing colour and texture features, some misclassifications persisted due to class imbalance and limited dataset size.

A study by Sarki et al. in 2021 applied ResNet50, VGG-16, Xception, as well as a new CNN model to classify image eye diseases. The dataset used consist of images of diabetic retinopathy, diabetic macular oedema, and Glaucoma. The study is structured into 2 stages, the first stage is using the pre-trained network on the raw dataset and the best model will be used later (VGG16 on this case). The second stage is to preprocess the images first with image enhancement, segmentation, and augmentation, then the processed dataset is used on VGG16 (best model on raw dataset) and the proposed new CNN network. The result is the new CNN network outperforming the VGG16 on all the pre-processed datasets.

In a recent study, Akter et al. (2024) proposed a CNN-based approach for multiclass MRI classification. The model used are five pre-trained CNN models such as VGG16, VGG19, EfficientNet-B0, EfficientNet-B7, and ResNet152V2, alongside with a newly developed model. The proposed CNN model achieved superior performance, surpassing all five pre-trained models which indicates its effectiveness in accurately classifying MRI.

Another study by Musthafa et al. (2024) focused on classifying lung cancer using a double-layered convolutional neural network. This research used the IQ-OTHNCCD lung cancer dataset, which consist of malignant, benign, and normal lung cancer CT scans. The model was built on a base CNN, incorporating SMOTE to address class

imbalance in the dataset. The results demonstrated an impressive accuracy of 99.64%, with a minimum precision of 96.7%.

The study by Li et al. (2024) applies the Xception deep learning model to classify MRI images across four stages of Alzheimer's Disease (AD) such as mild dementia, moderate dementia, non-dementia, and very mild dementia. The dataset used Alzheimer's Dataset from Kaggle. The images were processed with data augmentation for training, while original images were reserved for validation and testing. The custom network used the Xception's depth wise separable convolutions to optimize parameter efficiency and included Dropout layers to prevent overfitting, Dense layers with ReLU, and a SoftMax output for multi-class classification, using the Adamax optimizer and categorical cross-entropy as the loss function. The model achieved an impressive 99.6% accuracy, with high precision, recall, and F1 scores across categories.

The recent study by Yue and Li in 2024 introduces MedMamba, a hybrid model for medical image classification that combines convolutional layers with state space models (SSMs) to effectively capture local and long-range dependencies in diverse medical imaging tasks. MedMamba was evaluated on 16 datasets across ten imaging modalities totalling up to 411,007 images. Public and private collections like PAD-UFES-20 for skin lesions and CPN X-ray for COVID-19 and pneumonia detection was also included. The model employs a unique SS-Conv-SSM block, it integrates SSMs for long-range context and grouped convolution for local feature extraction, alongside with a channel-shuffle and 2D-selective-scan (SS2D) mechanism to enhance global context modelling with efficient computation. Three MedMamba variants (Tiny, Small, and Base) showed high performance and parameter efficiency. For instance, MedMamba-S achieved a 97.3% overall accuracy on the CPN X-ray dataset which surpasses baseline models with fewer parameters.

In 2024, Moldovanu and peers proposed a hybrid model combining CNNs, machine learning, and transfer learning to classify meningioma tumours and healthy brain tissue in MRI images. The dataset used is 3,264 MRI images from Kaggle. The study compares three CNN models which are an original CNN, DenseNet169, and EfficientNetV2B0, alongside with three ML classifiers: Random Forest (RF), K-Nearest Neighbors (KNN), and Support Vector Machine (SVM). They assessed the model's performance with metrics like accuracy and ROC-AUC. The EfficientNetV2B0-SVM combination achieved the highest classification accuracy of 99.5%.

In a 2024 study, Shouman and peers introduced a CNN-based method using DenseNet121 for classifying pelvic bone tumours. They seek to enhance diagnostic

precision in orthopaedic oncology. The approach used a publicly accessible pelvic CT dataset from Zenodo, consisting of 178 3D CT images sourced from CLINIC and CLINIC-metal sub-datasets, with diverse cases of benign and malignant pelvic tumours across 14 categories. The dataset included pre- and post-surgical scans that are gathered from multiple clinics and databases to ensure comprehensive coverage. The model DenseNet121 was chosen for its efficiency and feature propagation capabilities to effectively address challenges like the vanishing gradient problem and minimizing redundant mappings. The model was trained with a cross-entropy loss function and the Adam optimizer. The images were processed at a standardized resolution of (50, 50, 50) mm, with a batch size of 60 over 100 epochs. The model resulted in an impressive accuracy of 94%, with precision and recall scores above 90% across all tumour types.

In 2024, Ali and peers introduced a hybrid machine learning model designed to improve Alzheimer's disease detection through accurate segmentation and classification of MRI brain scans. Using approximately 6,400 T1-weighted MRI images from the ADNI database, classified into Normal, Mild, Moderate, and Very Mild stages, the study integrated Improved Fuzzy C-means (ImFCm) clustering, watershed segmentation, and a CNN-LSTM classifier. ImFCm was employed to segment brain tissue into grey matter, white matter, and cerebrospinal fluid, while watershed segmentation was applied twice to capture inner and outer brain features. The resulting segments were fed into a 14-layer CNN-LSTM network to classify AD severity levels. The model achieved 98.13% accuracy, with F1 scores around 98% across classes which outperforms traditional and state-of-the-art approaches.

In 2024, Preetha and peers used EfficientNet-B4 for detecting brain tumours in MRI images. The model was trained on the Brain Tumour Detection 2020 dataset from Kaggle (3,000 images divided into "tumour" and "non-tumour" classes). It achieved high accuracy through transfer learning and fine-tuning. Utilizing data augmentation and hyperparameter optimization via Bayesian Optimization, EfficientNet-B4 outperformed other models (e.g., VGG19, ResNet50, and other EfficientNet variants) in metrics such as accuracy (99.33%), F1-Score (99.34%), and recall (100%).

In a 2024 study by Rahman and peers, presented a lightweight CNN model GliomaCNN which aimed at classifying brain tumours in MRI scans into low-grade gliomas (LGG) and high-grade gliomas (HGG). The BraTS 2020 dataset that includes MRI scans of LGG and HGG tumours was used. The model incorporated a gradient-boosting approach for enhanced accuracy. As well as, used the explainable AI techniques, SHAP and Grad-CAM++, to provide interpretability. GliomaCNN achieved

a high classification accuracy of 99.1569%, which surpassed the other models while maintaining reduced computational complexity.

b. Previous Work in Medical Image Classification Using Visual Transformer Based Model

In 2023, Ren et al. proposed using Transformer for COVID19 medical image classification. The dataset used was 2 dataset of binary class lung CT scans and X-Ray that is separated to normal, infected virus, infected bacteria, and infected COVID19. The proposed model was a hybrid of ResNet and ViT, named RMT-Net. The proposed model resulted in a validation accuracy of 98.84% on the X-Ray dataset and 99.24% on the CT scans dataset, which outperforms ResNet-50, VGGNet-16, i-CapsNet and MGMADS-3.

Another study by Dai et al. in 2021 on Multi-Modal image Classification using Transformer-CNN hybrid network called TransMed. The model that was proposed, used CNN as low-level feature extractor for getting patch embedding that is inputted into the transformer encoder network. The dataset that was used are PGT Dataset that includes 344 head and neck MRI, with 2 modalities of MRI (T1 & T2), and MRNet Dataset that includes 1370 knee MRI examinations with each case labelled anterior cruciate ligament (ACL) tear, meniscus tear, or other signs of abnormalities in the knee. The proposed model resulted in it reaching the new state of the art for the 2 datasets. The TransMed-S reaching accuracy of 88.9% on the PGT Dataset, 91.8% on the abnormality MRNet dataset, 94.9% on the ACL tear MRNet dataset, and the TransMed-B reaching 85.3% on the meniscus tear MRNet dataset.

Another model was proposed by Okolo et al. in 2022 on chest X-ray image classification using transformer codenamed IEViT. The dataset used was chest X-Ray datasets of normal chest X-ray scans and pneumonia scans of children patients (1–5 years old), Tuberculosis Chest X-ray Database, The COVID-19 radiography database, and The COVIDx dataset. The proposed model resulted in better accuracy in all model size in comparison to the ViT model, with the best model IEViT-L/32 (IEViT-L with patch size of 32×32) 98.59%.

A study by Gai et al. (2023) compared CNN-based model and transformer-based model for identifying lung cancer, based on CT scans of the lung that are categorized as malignant and benign tumour. The model used is EfficientNet3D which is an incarnation of the EfficientNet model for processing 3-D medical images. The EfficientNet3D model was compared with Deit-S model (ViT in pytorch), which resulted in the EfficientNet3D

model achieving AUC score of 98.1% when pre-trained with SSL, outperforming DeiT-S model that only got 95.7% when pre-trained with SSL.

Chen et al. (2021) proposed a hybrid ConvNet-Transformer architecture for self-supervised volumetric image registration. This model uses ViT block to high-level features of moving and fixed images to learn long distance relationship between images. As evident from the results, the proposed ViT-V-Net achieved a significant gain of over 0.1 in Dice performance compared to other methods. As well as, a lower loss values and higher validation Dice scores during training. In conclusion, the ViT-based architecture demonstrated superior performance compared to the leading registration methods that proves the effectiveness of ViT-V-Net.

Other study that uses the ViT architecture is by Lu et al. (2024), it proposes a model codenames CTBViT to detect the presence of Tuberculosis in CT and X-Ray scans. The proposed model differs from the base ViT structure by introducing a patch-reduction-block (PRB) to improve the efficiency and training speed of the model and a randomized classifier to avoid overfitting. The dataset used was a private Tuberculosis dataset from Fourth People's Hospital of Huai'an City, Jiangsu Province, China. The dataset was separated into two classes which are DS (Drug Sensitive) Tuberculosis and MDR (Drug Resistant) Tuberculosis. The resulting model could accurately classify the presence of DS-TB and MDR-TB with precision of 97.6% and 99%, cumulative accuracy of 98.3%.

In the 2023 study by Manzari et al., they introduced MedViT which is a hybrid CNN-Transformer model that was developed to enhance the robustness and efficiency in medical image classification, particularly under adversarial conditions. The model was evaluated on the MedMNIST-V2 dataset, which is a collection of 12 biomedical image datasets spanning CT, X-ray, ultrasound, and OCT modalities across various classification tasks. MedViT combines convolutional blocks for local feature extraction with Transformer blocks for global feature processing. By using an Efficient Convolution Block (ECB), Local Transformer Block (LTB), and Transformer Augmentation Block (TAB) for robust performance. As well as, a new data augmentation method, Patch Momentum Changer (PMC), was introduced to improve adversarial resistance. MedViT achieved a high classification accuracy and AUC scores compared to previous CNN and AutoML models on the MedMNIST datasets which outperforms ResNet baselines in both classification accuracy and resilience against gradient-based adversarial attacks.

A study by Dixon et al. (2024) proposed a hybrid learning architecture that integrates vision transformers (ViT) and deep neural networks for brain tumor classification in MRI

scans. This architecture utilizes multiple feature types derived from MRI images such as texture features via GLCM and GLRLM descriptors, CNN-derived features for local pattern recognition, and ViT-derived features for capturing long-range dependencies. The model was tested on a combined dataset from Figshare, SARTAJ, and Br35H, as well as a local dataset. This hybrid model uses weighted feature fusion and a multi-layer perceptron (MLP) classifier. The results indicated high accuracy which outperforms models based solely on CNN or ViT. The system also showed strong generalization across different datasets by achieving over 99% accuracy, sensitivity, and specificity.

A study by Zhao et al. (2024) developed GMILT model to noninvasively predict EGFR mutation status in lung adenocarcinoma patients by using deep learning techniques on CT images. The dataset used consists of 512 patients and the model achieved an AUC of 0.772 on the internal test dataset, which outperforms traditional methods. The model integrates multiple instances learning and transformer architecture, utilizing invasiveness information to enhance prediction accuracy. Additionally, GMILT demonstrated great performance across external validation datasets and effectively identified the "core area" related to EGFR mutations, aiding in biopsy guidance.

The study by Xu and Meng in 2024 introduces Covid-ProtoPFormer as an interpretable vision transformer model designed to detect COVID-19 from chest medical images. The dataset used consists of chest X-ray and CT images. The model used a prototype block that enhances attention to core image content and replaces the last two layers of the transformer encoder. It achieved high classification accuracy of 99.3%, 96.8%, and 98.5% across different datasets. The loss function guides the learning of prototype features based on visual concepts, while the model's interpretability is enhanced through visualization techniques that reveal the reasoning process behind predictions. The results showed that the model not only excels in accuracy but also significantly improves interpretability.

A study by Uzuzaki et al. (2024) proposed Variable Vision Transformer (vViT) model to grade diffuse gliomas based on the 2021 WHO CNS tumor classification. By using the UCSF Preoperative Diffuse Glioma MRI (UCSF-PDGM) dataset, the study involved MRI scans from 501 patients with diffuse gliomas, classified into WHO grades 2, 3, and 4, with 1,575 MRI images per grade. The vViT model integrated multimodal data, including demographic information (age, sex), radiomic features from MRI (CET1, T2, FLAIR, and DWI), and MRI images segmented by tumor grade. The model excelled in multiclass classification with an accuracy of 84% and achieved its highest AUC-ROC

of 0.94 in binary classification between Grade 4 and Grades 2/3. Additionally, a Cohen's Kappa of 0.54 in multiclass grading highlighted moderate reliability.

Recent study by Wang et al. (2024) presents RanMerFormer, a vision transformer-based model designed to improve efficiency and accuracy in classifying brain tumors in MRIs. The model was evaluated on two public datasets which are DS1 with 7,023 images across four categories (glioma, meningioma, pituitary, and no tumor) and DS2 with 3,064 images for glioma, meningioma, and pituitary tumors. RanMerFormer combines a vision transformer (ViT) backbone with token merging to reduce redundant information, enhancing computational efficiency. An additional Random Vector Functional-Link (RVFL) head boosts the classification performance by using randomized features. The model achieved state-of-the-art results with 99.77% accuracy on DS1 and 98.86% on DS2 which outperforms existing methods.

Saha et al. (2024) present a Vision Transformer (ViT)-based model designed for identifying interstitial lung disease (ILD) patterns in chest CT scans. The model uses the MedGIFT dataset, consisting of 235 CT images across four classes: healthy, hypersensitivity pneumonitis, pulmonary fibrosis, and tuberculosis. By building upon DenseNet169 for feature extraction and enhancing with the ViT architecture, the model uses multi-head self-attention to capture spatial relationships within image patches. The proposed ViT-ILD achieved a training accuracy of 100%, validation accuracy of 98%, and test accuracy of 82.75%. Performance metrics for test data include recall at 74.15%, precision at 100%, and F1-score at 82.35%.

In 2024, Liu and Sun proposed a hybrid model, Reswin which combines ResNet-3D and Video Swin Transformer to classify COVID-19 cases in 3D CT images. Using the MosMedData dataset, the authors designed binary and triple classification tasks. In binary classification (COVID-19 vs. non-COVID), Reswin achieved 90.99% accuracy, 92.66% precision, and an F1-score of 94.25%. In triple classification (severe pneumonia, non-severe pneumonia, and healthy), it attained 86.55% accuracy and an F1-score of 86.2%.

Table 2.1 Summary of Literature Review Results on the Implementation of CNN and Transformer-Based Models in Medical Image Classification

Author & Year	Source	Objective	Method	Data	Conclusion
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Reddy et al. (2024)	Journal Article	Explored the efficacy of deep learning models for brain tumor detection and classification	ResNet-50, VGG16, Inception V3, and a custom CNN model	3,265 MRI images covering gliomas, meningiomas, pituitary tumors, and tumor-free images	The custom CNN model achieved superior performance, with a validation accuracy of 94.4%, an AUC of 97.53%, and a recall rate of 92.2%, outperforming the other architectures.
Yadav & Jadhav (2019)	Journal Article	The study aims to evaluate the performance of various methods like SVMs, CNNs, and capsule networks in diagnosing pneumonia from chest X-rays.	SVM classifier, VGG16 and InceptionV3	Chest X-ray images, which includes three categories: normal cases, bacterial pneumonia, and viral pneumonia.	The VGG16 model, fine-tuned with one retrained convolutional layer, achieved state-of-the-art results, emphasizing the importance of matching model complexity to dataset scale.
Gai et al. (2023)	Journal Article	Compares Vision Transformers (ViTs) and CNNs for lung cancer detection using CT scans.	Vision Transformers (ViTs), specifically Deit-S, and EfficientNet3D, which is a 3-D Convolutional Neural Network (CNN) model.	Lung cancer CT scan with 212 cases, consisting of 130 malignant and 82 benign tumor cases	CNNs outperformed Vision Transformers, achieving a recall of 93.4% and AUC of 98.1% with self-supervised learning, although Deit-S improved with transfer learning but still lagged behind CNNs.

Sarki et al. (2021)	Journal Article	The study seeks to improve automatic classification of diabetic eye diseases using CNNs and high-quality fundus images.	ResNet50, Xception, VGG16, and DenseNet21, as well as a newly built five-layered convolutional neural network (CNN) model	Diabetic Eye Diseases publicly accessible datasets, including Messidor, Messidor-2, DRISHTI-GS, and a Retinal Dataset from GitHub	Pre-trained CNN models with preprocessing techniques improved the classification accuracy of mild diabetic eye disease images, achieving up to 100% accuracy with VGG-16, although specificity was slightly below ideal.
Evwiekpaefe & Amrevuawho (2023)	Journal Article	Classifying dermatologic manifestations of cardiovascular disease using the EfficientNetV2 CNN model.	EfficientNetV2	dermatologic manifestations of cardiovascular disease	EfficientNetV2 achieved 97.98% training accuracy and 96.04% validation accuracy with 40 epochs, correctly classifying most images with minimal misclassifications.
Marques et al. (2020)	Journal Article	The study presents an automated COVID-19 diagnosis system using EfficientNet CNN on chest X-ray images, with binary and multi-class classification.	EfficientNet	chest X-ray images	In binary classification, the model achieved an average accuracy of 99.62%, recall of 99.63%, precision of 99.64%, and F1-score of 99.62%; for multi-class classification, it reached 96.70% accuracy, 96.69% recall, 97.54% precision, and 97.11% F1-score.

Nazir et al. (2022)	Journal Article	A hybrid CNN-Inception-V4 model is developed to classify breast cancer in mammography images, focusing on high accuracy and sensitivity.	CNN-Inception-V4, CNN, Inception-V4, and ResNet-50	CBIS-DDSM (Curated Breast Imaging Subset of DDSM) dataset	The CNN-Inception-V4 hybrid model achieved 99.2% accuracy, 99.8% sensitivity, 96.3% specificity, and 97% F1-score, outperforming models like CNN, Inception-V4, and ResNet-50.
Yubiao Yue & Zhenzhang Li (2024)	Journal Article	MedMamba is introduced as a novel architecture for medical image classification, combining CNNs and state space models (SSMs) for efficiency.	MedMamba, VMamba, Swin, ConvNeXt, ResNet18, and ResNet50	PathMNIST, DermaMNIST, OCTMNIST, Pneumonia MNIST, RetinaMNIST, BreastMNIST, BloodMNIST, OrganAMNIST, OrganCMNIST, and OrganSMNIST	MedMamba demonstrated competitive performance on most tasks with an efficient balance between model effectiveness and computational resource use.
Agarwal et al. (2023)	Journal Article	Developing an automated diagnostic model for Alzheimer's disease using EfficientNet.	EfficientNet-B0	458 MP-RAGE T1W MRIs, 229 of AD subjects and 229 of sMCI subjects	The EfficientNet-B0 model was implemented with 5-fold stratified cross-validation, achieving an average training accuracy of 95.29% across all folds.

Li et al. (2024)	Journal Article	Applies the Xception deep learning model to classify MRI images across four stages of Alzheimer's Disease (AD)	Xception	Brain MRI Scans of Alzheimer's Disease (AD)	The model achieved an impressive 99.6% accuracy, with high precision, recall, and F1 scores across categories.
Moldovanu et al. (2024)	Journal Article	Proposed a hybrid model combining CNNs, machine learning, and transfer learning to classify meningioma tumours and healthy brain tissue in MRI images.	original CNN, DenseNet169, EfficientNetV2B0, Random Forest (RF), K-Nearest Neighbors (KNN), and Support Vector Machine (SVM)	3,264 MRI images	EfficientNetV2B0-SVM combination achieved the highest classification accuracy of 99.5%, outperforming other combinations
Akter et al. (2024)	Journal Article	Proposed a CNN-based approach for multiclass MRI classification, utilizing five pre-trained CNN models such as VGG16, VGG19, EfficientNet-B0, EfficientNet-B7, and ResNet152V2, alongside with a newly developed model.	VGG16, VGG19, EfficientNet-B0, EfficientNet-B7, and ResNet152V2, alongside with a newly developed model.	6 types of Brain tumor MRI Image including four individual datasets and two merged sets, covering brain tumor types (glioma, meningioma, pituitary)	The CNN and U-Net-based models achieved high classification accuracy (up to 98.8%) in identifying brain tumor types from MRI scans, outperforming existing pre-trained models and showing potential for clinical application.

Mustha fa et al. (2024)	Journal Article	Classifying lung cancer using a double-layered convolutional neural network.	double- layered convolution al neural network	IQ- OTHNCCD lung cancer dataset	The results demonstrated a classification accuracy of 99.64%, with precision, recall, and F1-score values exceeding 98% across all categories.
Ren et al. (2023)	Journal Article	Proposed using hybrid of ResNet and ViT, named RMT-Net for COVID19 medical image classification	RMT-Net, ResNet-50, VGGNet- 16, i- CapsNet and MGMADS- 3	2 dataset of binary class lung CT scans and X- Ray that is separated to normal, infected virus, infected bacteria, and infected COVID19	Proposed model resulted in a validation accuracy of 98.84% on the X-Ray dataset and 99.24% on the CT scans dataset
Dai et al. (2021)	Journal Article	Proposed a Multi- Modal medical image Classification using Transformer- CNN hybrid network called TransMed	Transmed	PGT, MRNet, ACL tear MRNet, and the TransMed-B	The TransMed-S reaching accuracy of 88.9% on the PGT Dataset, 91.8% on the abnormality MRNet dataset, 94.9% on the ACL tear MRNet dataset, and the TransMed-B reaching 85.3% on the meniscus tear MRNet dataset.

Okolo et al. (2022)	Journal Article	Proposed a new transformer model codenamed IEViT on chest X-ray image classification.	IEViT	Chest X-Ray datasets of normal chest X-ray scans and pneumonia scans of children patients (1–5 years old), Tuberculosis Chest X-ray Database, The COVID-19 radiography database, and The COVIDx dataset	The proposed model resulted in better accuracy in all model size in comparison to the ViT model, with the best model IEViT-L/32 (IEViT-L with patch size of 32×32) 98.59%.
Chen et al. (2021)	Journal Article	Develop and evaluate ViT-V-Net, that combines Vision Transformer (ViT) and Convolutional Neural Networks (ConvNets) for unsupervised volumetric medical image registration	ViT-V-Net, VoxelMorph h-1, VoxelMorph h-2, NiftyReg, Symmetric Normalization (SyN)	260 T1-weighted brain MRI scans	ViT-V-Net achieved a significant gain of over 0.1 in Dice performance compared to other methods, along with lower loss values and higher validation Dice scores during training
Lu et al. (2024)	Journal Article	Proposes a model codenamed CTBViT to detect the presence of Tuberculosis in CT and X-Ray scans	CTBViT	Private TB dataset from Fourth People's Hospital of Huai'an City	The resulting model can accurately classify the presence of DS-TB and MDR-TB with precision of 97.6% and 99%, cumulative accuracy of 98.3%.

Manzari et al. (2023)	Journal Article	MedViT, a hybrid CNN-Transformer model, was developed to improve robustness and efficiency in medical image classification under adversarial conditions.	MedViT	MedMNIST-V2	MedViT achieves superior classification accuracy and AUC scores compared to previous CNN and AutoML models on the MedMNIST datasets.
Dixon et al. (2024)	Journal Article	Proposed a hybrid learning architecture that integrates vision transformers (ViT) and deep neural networks for enhanced brain tumor classification in MRI scans.	hybrid learning architecture that integrates vision transformers (ViT) and CNN	combined dataset from Figshare, SARTAJ, and Br35H, along with a local dataset	Results indicate high accuracy and improved robustness, outperforming models based solely on CNN or ViT.
Zhao et al. (2024)	Journal Article	Developed GMILT model to noninvasively predict EGFR mutation status in lung adenocarcinoma patients using deep learning techniques on CT images	GMILT	CT images comprise 512 patients	GMILT model achieved a high predictive accuracy for EGFR mutation status, with an AUC of 0.772 on internal tests and strong generalization on external datasets
Yang Xu & Zuqiang Meng (2024)	Journal Article	Introduces Covid-ProtoPFormer, an interpretable vision transformer model designed for detecting COVID-19 from chest medical images	Covid-ProtoPFormer	chest X-ray and CT images	The model achieved high classification accuracy of 99.3%, 96.8%, and 98.5% across different datasets

Uzuzaki et al. (2024)	Journal Article	Proposed Variable Vision Transformer (vViT) model for grading diffuse gliomas based on the 2021 WHO CNS tumor classification	vViT	the UCSF Preoperative Diffuse Glioma MRI (UCSF-PDGM)	The model excelled in multiclass classification with an accuracy of 84% and achieved its highest AUC-ROC of 0.94 in binary classification between Grade 4 and Grades 2/3.
Shouman et al. (2024)	Journal Article	Introduced a CNN-based method using DenseNet121 for classifying pelvic bone tumors, aiming to enhance diagnostic precision in orthopedic oncology	DenseNet121	pelvic CT dataset from Zenodo	The model resulted in an impressive accuracy of 94%, with precision and recall scores above 90% across all tumor types.
Ali et al. (2024)	Journal Article	Introduced a hybrid machine learning model to improve Alzheimer's disease detection through accurate segmentation and classification of MRI brain scans	CNN-LSTM with improved Fuzzy C-means and watershed segmentation techniques	6,400 T1-weighted MRI images from the ADNI database	The model achieved 98.13% accuracy, with F1 scores around 98% across classes, outperforming traditional and state-of-the-art approaches.
Wang et al. (2024)	Journal Article	Designed RanMerFormer, a vision transformer-based model to improve efficiency and accuracy in classifying brain tumors in MRIs	RanMerFormer	2 public brain MRI image datasets	The model achieved state-of-the-art results, with 99.77% accuracy on DS1 and 98.86% on DS2, outperforming other existing methods.

Saha et al. (2024)	Journal Article	Present a ViT-based model called ViT-ILD for identifying interstitial lung disease (ILD) patterns in chest CT scans.	ViT-ILD	MedGIFT	The proposed ViT-ILD achieved a training accuracy of 100%, validation accuracy of 98%, and test accuracy of 82.75%.
Liu & Sun (2024)	Journal Article	Propose a hybrid model, Reswin, combining ResNet-3D and Video Swin Transformer to classify COVID-19 cases in 3D CT images	Reswin	MosMedData	Reswin achieved 90.99% accuracy, 92.66% precision, and an F1-score of 94.25%. In triple classification (severe pneumonia, non-severe pneumonia, and healthy), it attained 86.55% accuracy and an F1-score of 86.2%.
Preetha et al. (2024)	Journal Article	Leverage EfficientNet-B4 for detecting brain tumors in MRI images.	EfficientNet-B4	Brain Tumor Detection 2020 dataset from Kaggle	EfficientNet-B4 outperformed other models in metrics such as accuracy (99.33%), F1-Score (99.34%), and recall (100%)
Rahman et al. (2024)	Journal Article	Presented a lightweight CNN model, GliomaCNN, aimed at classifying brain tumors in MRI scans into low-grade gliomas (LGG) and high-grade gliomas (HGG)	GliomaCNN	BraTS 2020	GliomaCNN achieved a high classification accuracy of 99.1569%, surpassing other models while maintaining reduced computational complexity.

2.2 Literature Review

2.2.1. Degenerative Lumbar Spine

Degenerative lumbar spine disease is a significant cause of disability worldwide. This disease includes various conditions such as spondylolisthesis, disc degeneration, and lumbar spinal stenosis. Patients often experience symptoms such as lower back pain (LBP), weakness, and lower extremity pain (Ravindra et al., 2018, p. 784). The degeneration process is categorized into three stages which are A, B, and C based on the progression's location and sequence (Kushchayev et al., 2018, p. 255).

Stage A affects the nucleus pulposus. Normally, the nucleus pulposus functions hydrostatically by transmitting pressure evenly to the surrounding annulus fibrosus and endplates based on Pascal's principle. Degeneration leads to dehydration of the nucleus pulposus, reducing intradiscal pressure. This places more mechanical load on the annulus fibrosus, which can lead to cracks and cavities that may progress to clefts and fissures (Kushchayev et al., 2018, p. 255). The progression of Stage A changes is often evaluated with the Pfirrmann grading system. Phenomena such as the vacuum effect, intradiscal fluid accumulation, and calcification may also occur (Kushchayev et al., 2018, pp. 254-256).

Stage B involves changes in the annulus fibrosus, endplates, and bone marrow. In a healthy state, the annulus fibrosus appears as a hypointense signal on MRI. Degenerative changes include circumferential, peripheral rim, or radial fissures, and disk displacement beyond the intervertebral space limits, categorized as bulging or focal herniation. Herniations can appear as anterior or posterior in the axial plane and are classified further as central, paracentral, foraminal, or paraforaminal. Endplate changes or damage is a distinctive marker of degeneration and is classified on MRI from Type 1 (normal) to Type 6 (extensive endplate damage). Bone marrow changes include Type 1 corresponds to bone marrow oedema and vascularized fibrous tissue, which may also be chronic or acute and strongly correlated to lumbar pain or instability; Type 2 reflects the presence of yellow bone marrow in the bone; and Type 3 that represents dense woven bone with no bone marrow, often stable and asymptomatic. Intravertebral instability is categorized into axial/vertical instability which typically caused by focal or diffuse pathology, and horizontal instability, a reduction in the ability of discs, facet joints, and ligaments to maintain alignment. Additional conditions include degenerative and cervical spondylolisthesis and spondylosis (Kushchayev et al., 2018, pp. 256-265).

Stage C primarily impacts facet joints, ligamentum flavum, and the spinal canal. Facet joints that found at every spinal level except C1-C2 are true synovial joints and may develop osteoarthritis, which can occur independently or because of disc degeneration and disc space narrowing. This condition can be assess using the 4-tiered grading scale. Ligamentum flavum thickening often correlates with disc degeneration and herniation. Spinal canal stenosis, classified as central, lateral, or foraminal, is assessed by several grading systems. The Kang grading system rates cervical spinal canal stenosis from Grade 0 that indicates no stenosis to Grade 3 that indicates elevated T2 MRI signal indicating severe cord compression. The Park grading system evaluates cervical foraminal stenosis from Grade 0 indicating no stenosis to Grade 3 indicating the nerve root morphological change). Lumbar central stenosis is assessed by the Bartynski system, and foraminal stenosis by the Wildermuth classification (Kushchayev et al., 2018, pp. 265-270).

2.2.2. Transformer

The Transformer architecture was first introduced by Vaswani et al. in 2017. It was developed with only using attention mechanisms to capture the global features between input and output. The architecture of the base transformer itself, rely on a stack of encoder and decoder block. The encoder stack consists of $N = 6$ identical encoder layers, with each layer consist of a multi-head self-attention mechanism and a position-wise feed-forward network, with a residual connection around each of the sub layer and a bath normalization after each sub layer. The output of each sub layer will be $LayerNorm(x + Sublayer(x))$. With each $Sublayer(x)$ are function for multi-head self-attention mechanism and a position-wise feed-forward network. Resulting in an output dimension of $d_{model} = 512$.

The decoder stack also consists of $N = 6$ identical decoder layer, with each layer consisting of the same 2 sub layers as the encoder stack. However, each decoder layer includes an additional multi-head attention mechanism that attends to the encoder's outputs. The self-attention layer of the decoder stack is also modified by preventing the position from attending to subsequent positions, with offsetting the output embedding by 1 position. This is done to ensures that the prediction of position i is only predicted by positions less than i . The novel approach of only relying on attention mechanisms, resulted in the base Transformer model establishing the new state-of-the-art *BLEU* score of 28.4 on WMT 2014 English-to-German translation task (Vaswani et al., 2017).

In practical applications, the Transformer model processes sequences by encoding words into Word Embeddings (WE) and decoding them into words at the output (Santana

et al., 2022, p. 37). Despite its success, the Transformer model is computationally demanding due to the high memory and time complexity associated with self-attention and cross-attention layers (Zhu et al., 2021, p. 2). Another significant drawback is factual hallucination, particularly in Large Language Models (LLMs), where a lack of intrinsic knowledge can lead to incorrect or misleading outputs (Guan et al., 2023, p. 1).

The success of Transformers has revolutionized natural language processing (NLP) and has expanded to computer vision, with models like the Vision Transformer (ViT) achieving state-of-the-art performance on image classification through a fully attention-based framework (Zheng et al., 2021, pp. 6882-6883). Furthermore, Transformers distinguish themselves by their ability to handle the 2-matching problem, which feed-forward neural networks are unable to solve (Peng et al., 2024, p. 1). This versatility is due to the model's robust capacity for handling long-term dependencies in text, surpassing the limitations of Recurrent Neural Networks (RNNs) (Radford et al., 2018, p. 2).

2.2.3. Convolutional Neural Network (CNN)

Convolutional Neural Network (CNN) is an advanced deep learning architecture, that has been specialized to handle multi-dimensional data structures effectively and automatically (e.g image) (Li et al., 2024, p. 3). CNN has been a revolution in computer vision since 2012, It was first introduced in its participation of ImageNet Large Scale Visual Recognition Challenge (ILSVRC). CNN is also the current state-of-the-art architecture in many computers vision task (Rangel et al, 2024).

CNN is comprised of neurons that is separated to 3 dimensions, the height, width, and depth. For example, if given an image of size $64 \times 64 \times 3$, the output will be $1 \times 1 \times n$ (n being the number of class). The overall architecture of CNN itself, consist of Convolution, Pooling, and Fully Connected Layers (O'shea & Nash, 2015). CNN distinguishes itself by effectively extracting local features, but it also has a difficulty in capturing global features and long-range tendencies (Yue & Li, 2024, p. 1).

The success of CNN or Convolutional Networks has been attributed to the presence of large public image repositories, such as ImageNet, and the new presence of high-performance computing system, like GPU (Graphical Proccession Unit) (Simonyan & Zisserman, 2015). The attractiveness of CNN is their ability to be controlled by varying the depth and breadth of the model. CNN also have much fewer connection in comparison to other feed-forward network, which made it easier to train (Krizhevsky et al., 2017).

2.2.4. Models

c. Inception-V4

Inception-V4 was introduced in 2016 by Christian Szegedy et al. in their paper "Inception-V4, Inception-ResNet and the Impact of Residual Connections on Learning." It builds upon the Inception deep convolutional architecture, which was initially introduced in 2014 with the paper "Going Deeper with Convolutions," codenamed Inception-V1. Inception-V1 introduced the concept of using multiple filter sizes in parallel, enabling the network to capture various visual features at different scales.

Inception-V4 incorporates three types of Inception blocks which are A, B, and C each carefully optimized for different operations to handle various feature extraction tasks effectively. Additionally, it includes reduction blocks specifically designed to adjust the width and height of feature maps, enhancing the network's flexibility in capturing complex patterns. The architecture also expands filter banks to alleviate representation bottlenecks, which improves the model's learning capacity and efficiency. In experimental evaluations, Inception-V4 achieved a top-5 error rate of 3.08%, comparable to the more computationally expensive Inception-ResNet-v2 model, demonstrating its efficiency and impressive performance (Szegedy et al., 2016).

d. EfficientNet

In 2019, Mingxing Tan and Quoc V. Le introduced EfficientNet, a series of convolutional neural network models that significantly advanced the field of image classification. The largest model, EfficientNet-B7, achieved a remarkable 84.3% top-1 accuracy, establishing a new state-of-the-art performance at the time while being 8.4 times smaller and 6.4 times faster than GPipe, the previous benchmark model. Even the smallest variant, EfficientNet-B0, outperformed models with 4.9 times more parameters and was 11 times faster in execution.

The exceptional performance of EfficientNet was realized through a technique known as neural architecture search, which was influenced by earlier work in the 2016 paper "Neural Architecture Search with Reinforcement Learning" by Barret Zoph and Quoc V. Le. This approach allowed for the development of a baseline model optimized for floating point operations per second (FLOPS). EfficientNet-B0, the resulting baseline model, consists of convolutional blocks with the mobile inverted bottleneck (MBConv) serving as its primary building block. To enhance performance

further, a compound scaling method was subsequently applied to EfficientNet-B0, resulting in the creation of models ranging from EfficientNet-B1 to EfficientNet-B7.

e. EfficientNetV2

EfficientNetV2 was first introduced in 2021 by Mingxing Tan and Quoc V. Le in the study “EfficientNetV2: Smaller Models and Faster Training”. EfficientNetV2 outperforms other models significantly, with using less parameter and faster training time, up to 11x faster and 6.8x less parameter. This great efficiency and performance can be contributed to the model approach using training-aware Neural Architecture Search (NAS) and scaling (Tan & Le, 2021, p. 2).

The EfficientNetV2’s NAS network draws major influence from the earlier MnasNet model by Tan et al. (2018) and the EfficientNet framework developed by Tan and Le (2019) with the primary objective to optimize model accuracy, parameter efficiency, and training efficiency, particularly on modern hardware. The EfficientNetV2 search space closely resembles the Factorized Hierarchical Search Space used in MnasNet, with key modifications of reducing the search space. This refinement was achieved by removing search options deemed unnecessary for EfficientNetV2’s structure and reusing channel sizes from the base model EfficientNet. For larger model variations, reinforcement learning, and random search methods were employed to explore optimal configurations while maintaining computational efficiency (Tan & Le, 2021, p. 4).

The architecture of EfficientNetV2 was based upon the original EfficientNet design with several key modifications. EfficientNetV2 extensively utilizes MBConv and fused-MBConv blocks inspired by EfficientNet-EdgeTPU in its early layers. Additionally, it applies smaller expansion ratios within the MBConv layers and reduces kernel sizes to 3×3 , while compensating with an increased number of layers. Lastly, EfficientNetV2 omits the final stride-1 stage present in the base EfficientNet model, enhancing efficiency and performance (Tan & Le, 2021, p. 3).

f. Vision Transformer (ViT)

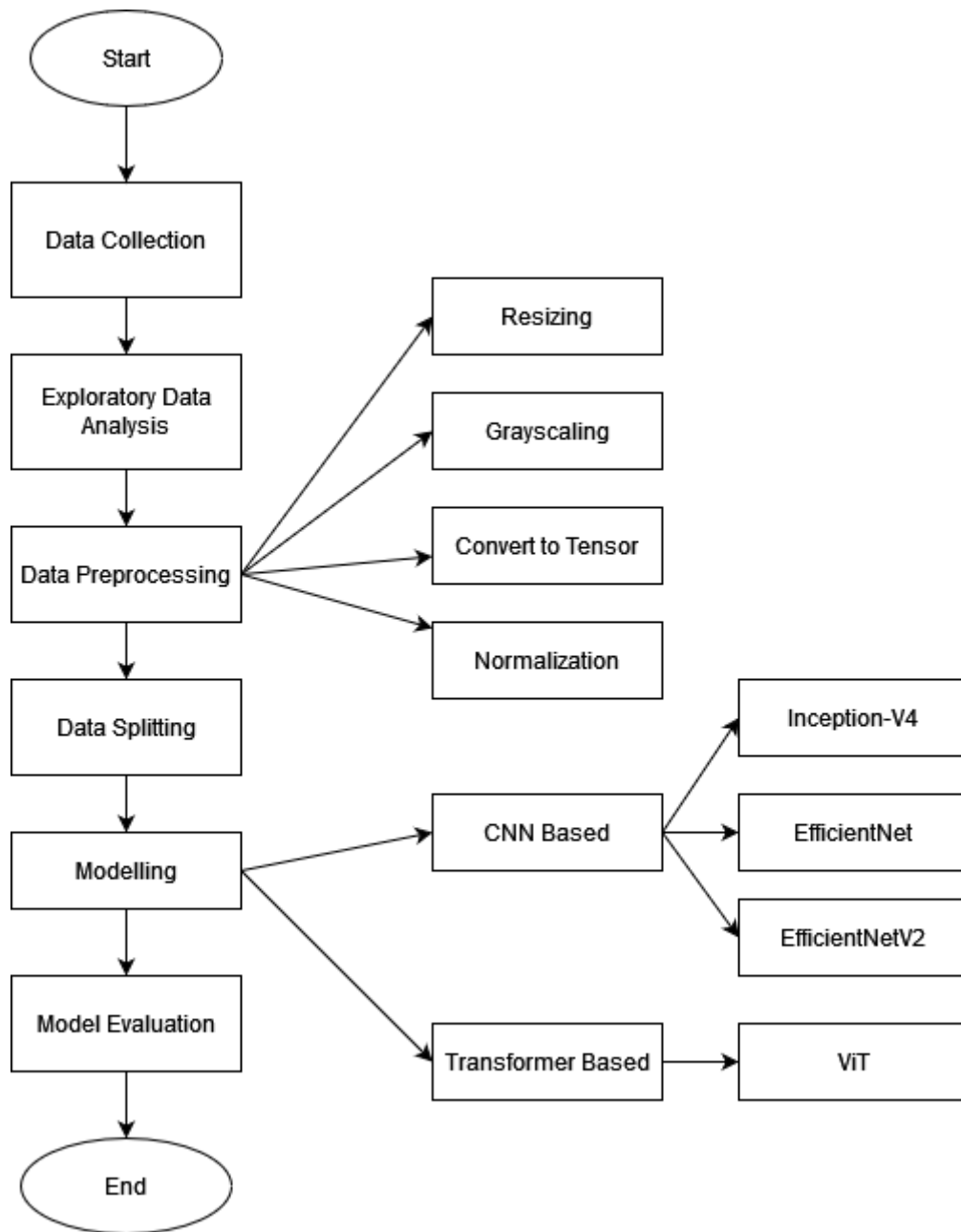
Vision Transformer or ViT was introduced in 2021 by Dosovitskiy et al., which was contributed to by the success of Transformer model in natural language processing (NLP) tasks. ViT works by treating images like a sentence or documents. It works by splitting an image to patches, with the sequence of linear embedding of the patches as the input for the transformer. Architecturally ViT strongly followed the

original Transformer model, however, it only uses the encoder stack. The resulted models are ViT-Base, ViT-Large, and ViT-Huge with the difference being the number of heads in each model of 12, 16, and 16 respectively and number of layers of 12, 24, and 32, respectively. This approach resulted in the ViT-L/16 (ViT-L with patch size of 16×16) outperforming the ResNet152x4, with also being faster.

CHAPTER 3

IMPLEMENTATION METHOD

3.1 Thinking Framework



Picture 1 – Methodology Flowchart

3.2 Research Methods

In this study, the image classification task follows a structured pipeline that begins with data collection, where images relevant to the classification problem are gathered. Following

this, exploratory data analysis (EDA) is conducted to assess the dataset, to see the distribution of classes, image quality, and identifying any initial patterns that may inform preprocessing decisions. The data preprocessing stage includes several key steps, such as resizing (to standardize image dimensions), grayscaling (converting images to grayscale), tensor conversion (transforming the data into tensor format for model compatibility), and normalization (scaling pixel values for consistency across all images).

The data is then split into training, validation, and testing sets to facilitate model development. During the modelling phase, two types of models are trained: CNN-based models, which include architectures like Inception-V4, EfficientNet, and EfficientNetV2; and Transformer-based models, specifically using the Vision Transformer (ViT) architecture. These models are evaluated and compared to identify the most effective approach for classifying degenerative lumbar spine conditions. The evaluation phase measures model performance using accuracy as the primary metric, allowing for a straightforward assessment of each model's classification accuracy and guiding the selection of the optimal architecture for this task.

3.2.1 Data Collection

The dataset used in this study, titled "RSNA 2024 Lumbar Spine Degenerative Classification," was sourced from the official Kaggle platform. Curated and annotated by the Radiological Society of North America (RSNA) and the American Society of Neuroradiology (ASNR), it contains MRI scans labelled with severity levels (Normal/Mild, Moderate, or Severe) for five degenerative lumbar spine conditions across intervertebral disc levels from L1/L2 to L5/S1. The conditions include Left Neural Foraminal Narrowing, Right Neural Foraminal Narrowing, Left Subarticular Stenosis, Right Subarticular Stenosis, and Spinal Canal Stenosis. The dataset was sourced from multiple international institutions, providing a diverse and representative sample of degenerative spine conditions.

3.2.2 Exploratory Data Analysis

The dataset comprises five primary categories relating to spinal abnormalities, each labeled according to severity: "Normal/Mild," "Moderate," and "Severe." The categories and their distribution across severity levels are as follows:

1. Spinal Canal Stenosis

This category contains a total of 9,753 samples, with a heavy skew toward the "Normal/Mild" label (8,552 samples, representing approximately 87.7%). The "Moderate" and "Severe" classes contain significantly fewer samples, at 732 (7.5%)

and 469 (4.8%) respectively. This distribution indicates an imbalance that may require special attention to ensure fair model performance across all severity levels.

2. Left Neural Foraminal Narrowing

Comprising 9,860 samples, this category also displays a majority of "Normal/Mild" cases, totaling 7,671 (77.8%), followed by 1,792 "Moderate" cases (18.2%) and 397 "Severe" cases (4.0%). The noticeable disparity among these classes suggests potential challenges for classification models, especially in accurately identifying the "Severe" cases due to their limited representation.

3. Right Neural Foraminal Narrowing

This group includes 9,829 samples, labeled as "Normal/Mild" (7,684 samples, or 78.2%), with 1,767 samples labeled "Moderate" (18.0%) and 378 as "Severe" (3.8%). Like the left neural foraminal narrowing, this dataset shows an imbalance, which may influence model sensitivity towards underrepresented classes.

4. Left Subarticular Stenosis

With a total of 9,603 samples, this category is also skewed toward the "Normal/Mild" label, with 6,857 samples (71.4%), while "Moderate" cases represent 19.1% (1,834 samples) and "Severe" cases make up 9.5% (912 samples). This distribution indicates that while "Severe" cases are more prominent than in other categories, they remain underrepresented.

5. Right Subarticular Stenosis

This category, totaling 9,612 samples, reflects a similar imbalance, with "Normal/Mild" cases forming the majority at 6,862 samples (71.4%). The "Moderate" class contains 1,825 samples (19.0%), and the "Severe" class has 925 samples (9.6%). As with the left subarticular stenosis, this distribution may affect the model's capacity to generalize well across severity levels.

Based on the data described, "Normal/Mild" samples dominate, while "Severe" cases are markedly underrepresented. This class imbalance is a critical consideration, as it can impact model performance, potentially leading to biased predictions that may overlook "Severe" cases.

3.2.3 Data Preprocessing

Dataset that has been previously acquired cannot be directly employed for training and testing models, it must first undergo a preprocessing stage. Data preprocessing is an essential process involves preparing raw data and transforming it into a format suitable for the intended

task. Preprocessing step itself is needed to eliminate noise/variation, as well as improving the quality and contrast of the image which can affect the accuracy of the model itself (Sarki et al., 2021, p. 458). The data preprocessing steps taken in this study are as follows:

- a. Resizing the image into the dimension of 128×128 pixels
- b. Grayscale the image to reduce the colour information to a single channel, which simplifies the data.
- c. Convert the data to a PyTorch tensor. This conversion not only transforms the data into a format compatible with PyTorch but also rescales the pixel values from the original range of $[0, 255]$ (for uint8 images) to a normalized range of $[0.0, 1.0]$ (for float tensors).
- d. Normalize the pixel values using specific mean and standard deviation values. This process will help in model training, as it ensures that the input data is consistent and comparable to the data the model was originally trained on.

3.2.4 Data Splitting

After preprocessing, the training data will be split into a 70:30 ratio, allocating 70% for training and 30% for validation. The validation set is crucial for these complex models to fine-tune hyperparameters, prevent overfitting, and ensure better generalization (Vaswani et al., 2017). Regarding the test data, it has been provided by RSNA, ensuring a standardized evaluation of the model's performance.

3.2.5 Modelling

During the modelling phase, two types of models are trained: CNN-based models, which include architectures such as Inception-V4, EfficientNet, and EfficientNetV2, and lightweight transformer models, specifically utilizing the Vision Transformer (ViT) architecture. This dual approach allows for leveraging the strengths of both convolutional neural networks and transformer models to enhance performance in the classification of degenerative lumbar spine conditions.

3.2.6 Model Evaluation

After obtaining results from the testing stage, it is essential to evaluate the effectiveness of each model. Accuracy will serve as the primary metric for this evaluation process. To calculate accuracy, we need to determine the values of True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN) (Ichsan et al., 2022, p. 609). Here, TP represents the number of data points correctly classified as positive, FP indicates the number of data points

incorrectly classified as positive, TN refers to the number of data points accurately classified as negative, and FN denotes the number of data points inaccurately classified as negative.

Accuracy is an evaluation metric that assesses how well a classification model correctly predicts cases in a dataset. This metric indicates the model's overall ability to make accurate predictions. The following is the formula for calculating the accuracy score (Ichsan et al., 2022, p. 609):

$$\textit{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

REFERENCES

- Agarwal, D., Berbís, M. Á., Luna, A., Lipari, V., Ballester, J. B., & de la Torre-Díez, I. (2023). Automated Medical Diagnosis of Alzheimer's Disease Using an Efficient Net Convolutional Neural Network. *Journal of Medical Systems*, 47(1), 57. <https://doi.org/10.1007/s10916-023-01941-4>
- Akter, A., Nosheen, N., Ahmed, S., Hossain, M., Yousuf, M. A., Almoyad, M. A. A., Hasan, K. F., & Moni, M. A. (2024). Robust clinical applicable CNN and U-Net based algorithm for MRI classification and segmentation for brain tumor. *Expert Systems with Applications*, 238, 122347. <https://doi.org/10.1016/j.eswa.2023.122347>
- Ali, E. H., Sadek, S., El Nashef, G. Z., & Makki, Z. F. (2024). Advanced Integration of Machine Learning Techniques for Accurate Segmentation and Detection of Alzheimer's Disease. *Algorithms*, 17(5), Article 5. <https://doi.org/10.3390/a17050207>
- Bazi, Y., Bashmal, L., Rahhal, M. M. A., Dayil, R. A., & Ajlan, N. A. (2021). Vision Transformers for Remote Sensing Image Classification. *Remote Sensing*, 13(3), Article 3. <https://doi.org/10.3390/rs13030516>
- Chen, J., He, Y., Frey, E. C., Li, Y., & Du, Y. (2021). ViT-V-Net: Vision Transformer for Unsupervised Volumetric Medical Image Registration (No. arXiv:2104.06468). arXiv. <http://arxiv.org/abs/2104.06468>
- Chen, X., Wang, X., Zhang, K., Fung, K.-M., Thai, T. C., Moore, K., Mannel, R. S., Liu, H., Zheng, B., & Qiu, Y. (2022). Recent Advances and Clinical Applications of Deep Learning in Medical Image Analysis. *Medical Image Analysis*, 79, 102444. <https://doi.org/10.1016/j.media.2022.102444>
- Dai, Y., Gao, Y., & Liu, F. (2021). TransMed: Transformers Advance Multi-Modal Medical Image Classification. *Diagnostics*, 11(8), Article 8. <https://doi.org/10.3390/diagnostics11081384>
- Dixon, J., Akinniyi, O., Abdelhamid, A., Saleh, G. A., Rahman, M. M., & Khalifa, F. (2024). A Hybrid Learning-Architecture for Improved Brain Tumor Recognition. *Algorithms*, 17(6), Article 6. <https://doi.org/10.3390/a17060221>
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2021). An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale (No. arXiv:2010.11929). arXiv. <http://arxiv.org/abs/2010.11929>
- Evwiekpaefe, A. E., & Amrevuawho, O. F. (2023). CLASSIFICATION OF DERMATOLOGIC MANIFESTATIONS OF CARDIOVASCULAR DISEASE USING EFFICIENTNETV2 CNN MODEL. *International Journal of Intelligent Computing and Information Sciences*, 23(1), 115–127. <https://doi.org/10.21608/ijicis.2023.184311.1242>
- Gai, L., Xing, M., Chen, W., Zhang, Y., & Qiao, X. (2023). Comparing CNN-based and transformer-based Models for Identifying Lung cancer: Which Is More effective?

- Multimedia Tools and Applications*, 83(20), 59253–59269.
<https://doi.org/10.1007/s11042-023-17644-4>
- Guan, X., Liu, Y., Lin, H., Lu, Y., He, B., Han, X., & Sun, L. (2023). *Mitigating Large Language Model Hallucinations via Autonomous Knowledge Graph-based Retrofitting* (No. arXiv:2311.13314). arXiv. <http://arxiv.org/abs/2311.13314>
- Ichsan, M. N., Armita, N., Minarno, A. E., Sumadi, F. D. S., & Hariyady. (2022). Increased Accuracy on Image Classification of Game Rock Paper Scissors using CNN. *Jurnal RESTI (Rekayasa Sistem Dan Teknologi Informasi)*, 6(4), Article 4. <https://doi.org/10.29207/resti.v6i4.4222>
- Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2017). ImageNet classification with deep convolutional neural networks. *Communications of the ACM*, 60(6), 84–90. <https://doi.org/10.1145/3065386>
- Kushchayev, S. V., Glushko, T., Jarraya, M., Schuleri, K. H., Preul, M. C., Brooks, M. L., & Teytelboym, O. M. (2018). ABCs of the degenerative spine. *Insights into Imaging*, 9(2), 253–274. <https://doi.org/10.1007/s13244-017-0584-z>
- Li, S., Qu, H., Dong, X., Dang, B., Zang, H., & Gong, Y. (2024). *Leveraging Deep Learning and Xception Architecture for High-Accuracy MRI Classification in Alzheimer Diagnosis* (No. arXiv:2403.16212). arXiv. <http://arxiv.org/abs/2403.16212>
- Liu, C., & Sun, C. (2024). A Fusion Deep Learning Model of ResNet and Vision Transformer for 3D CT Images. *IEEE Access*, 12, 93389–93397. IEEE Access. <https://doi.org/10.1109/ACCESS.2024.3423689>
- Lu, S.-Y., Zhu, Z., Tang, Y., Zhang, X., & Liu, X. (2025). CTBViT: A novel ViT for tuberculosis classification with efficient block and randomized classifier. *Biomedical Signal Processing and Control*, 100, 106981. <https://doi.org/10.1016/j.bspc.2024.106981>
- MAGAYANE, D. A. (2021). *Clinical Classification and Risk of Chronification among Patients Presenting with Low Back Pain at Tertiary Care Level in Tanzania*. <http://ir.jkuat.ac.ke/handle/123456789/5559>
- Manzari, O. N., Ahmadabadi, H., Kashiani, H., Shokouhi, S. B., & Ayatollahi, A. (2023). MedViT: A robust vision transformer for generalized medical image classification. *Computers in Biology and Medicine*, 157, 106791. <https://doi.org/10.1016/j.compbiomed.2023.106791>
- Marcellino, C., Rabinstein, A. A., Liebo, G. B., Maus, T. P., & Bartleson, J. D. (2021). *Low Back Pain Emergencies* (2nd ed., p. 73). Springer, Cham. https://doi.org/10.1007/978-3-030-75778-6_2
- Marques, G., Agarwal, D., & de la Torre Díez, I. (2020). Automated medical diagnosis of COVID-19 through EfficientNet convolutional neural network. *Applied Soft Computing*, 96, 106691. <https://doi.org/10.1016/j.asoc.2020.106691>
- Moldovanu, S., Tăbăcaru, G., & Barbu, M. (2024). Convolutional Neural Network–Machine Learning Model: Hybrid Model for Meningioma Tumour and Healthy Brain Classification. *Journal of Imaging*, 10(9), Article 9. <https://doi.org/10.3390/jimaging10090235>

- Musthafa, M. M., Manimozhi, I., Mahesh, T. R., & Guluwadi, S. (2024). Optimizing double-layered convolutional neural networks for efficient lung cancer classification through hyperparameter optimization and advanced image pre-processing techniques. *BMC Medical Informatics and Decision Making*, 24(1), 142. <https://doi.org/10.1186/s12911-024-02553-9>
- Nazir, M. S., Khan, U. G., Mohiyuddin, A., Al Reshan, M. S., Shaikh, A., Rizwan, M., & Davidekova, M. (2022). A Novel CNN-Inception-V4-Based Hybrid Approach for Classification of Breast Cancer in Mammogram Images. *Wireless Communications and Mobile Computing*, 2022(1), 5089078. <https://doi.org/10.1155/2022/5089078>
- Okolo, G. I., Katsigiannis, S., & Ramzan, N. (2022). IEViT: An enhanced vision transformer architecture for chest X-ray image classification. *Computer Methods and Programs in Biomedicine*, 226, 107141. <https://doi.org/10.1016/j.cmpb.2022.107141>
- O'shea, K., & Nash, R. (2015). *An Introduction to Convolutional Neural Networks*. <https://arxiv.org/pdf/1511.08458>
- Park, C.-H., Min, K.-B., Min, J.-Y., Kim, D. H., Seo, K. M., & Kim, D.-K. (2021). Strong Association of Type 2 Diabetes with Degenerative Lumbar Spine Disorders. *Scientific Reports*, 11(1). <https://doi.org/10.1038/s41598-021-95626-y>
- Patil, S., & Gondhali, G. (2021). COVID-19 Pneumonia with Pulmonary Tuberculosis: Double Trouble. *The International Journal of Mycobacteriology*, 10(2), 206. https://doi.org/10.4103/ijmy.ijmy_51_21
- Peng, B., Narayanan, S., & Papadimitriou, C. (2024). *On Limitations of the Transformer Architecture* (No. arXiv:2402.08164). arXiv. <http://arxiv.org/abs/2402.08164>
- Preetha, R., Priyadarsini, M. J. P., & Nisha, J. S. (2024). Automated Brain Tumor Detection from Magnetic Resonance Images Using Fine-Tuned EfficientNet-B4 Convolutional Neural Network. *IEEE Access*, 12, 112181–112195. IEEE Access. <https://doi.org/10.1109/ACCESS.2024.3442979>
- Radford, A., Narasimhan, K., Salimans, T., & Sutskever, I. (2018). *Improving Language Understanding by Generative Pre-Training*. https://cdn.openai.com/research-covers/language-unsupervised/language_understanding_paper.pdf
- Rahman, M., Masum, M., Hasib, K., Mridha, M., Alfarhood, S., Safran, M., & Che, D. (2024). GliomaCNN: An Effective Lightweight CNN Model in Assessment of Classifying Brain Tumor from Magnetic Resonance Images Using Explainable AI. *Computer Modeling in Engineering & Sciences*, 140(3), 2425–2448. <https://doi.org/10.32604/cmes.2024.050760>
- Rangel, G., Cuevas-Tello, J. C., Nunez-Varela, J., Puente, C., & Silva-Trujillo, A. G. (2024). A Survey on Convolutional Neural Networks and Their Performance Limitations in Image Recognition Tasks. *Journal of Sensors*, 2024(1). <https://doi.org/10.1155/2024/2797320>
- Ravindra, V. M., Senglaub, S. S., Rattani, A., Dewan, M. C., Härtl, R., Bisson, E., Park, K. B., & Shrimel, M. G. (2018). Degenerative Lumbar Spine Disease: Estimating Global Incidence and Worldwide Volume. *Global Spine Journal*, 8(8), 784–794. <https://doi.org/10.1177/2192568218770769>

- Reddy, L. C. S., Elangovan, M., Vamsikrishna, M., & Ravindra, C. (2024). Brain Tumor Detection and Classification Using Deep Learning Models on MRI Scans. *EAI Endorsed Transactions on Pervasive Health and Technology*, 10. <https://doi.org/10.4108/eetpht.10.5553>
- Ren, K., Hong, G., Chen, X., & Wang, Z. (2023). A COVID-19 medical image classification algorithm based on Transformer. *Scientific Reports*, 13(1), 5359. <https://doi.org/10.1038/s41598-023-32462-2>
- Saha, S., Kumar, A., & Nandi, D. (2024). ViT-ILD: A Vision Transformer-based Neural Network for Detection of Interstitial Lung Disease from CT Images. *Procedia Computer Science*, 235, 779–788. <https://doi.org/10.1016/j.procs.2024.04.074>
- Santana, I. N., Oliveira, R. S., & Nascimento, E. G. S. (2022). Text Classification of News Using Transformer-based Models for Portuguese. *Journal of Systemics, Cybernetics and Informatics*, 20(5), 33–59. <https://doi.org/10.54808/JSCI.20.05.33>
- Sarki, R., Ahmed, K., Wang, H., Zhang, Y., Ma, J., & Wang, K. (2021). Image Preprocessing in Classification and Identification of Diabetic Eye Diseases. *Data Science and Engineering*, 6(4), 455–471. <https://doi.org/10.1007/s41019-021-00167-z>
- Shouman, M., Rahouma, K. H., & Hamed, H. F. A. (2024). A deep learning-based system for accurate diagnosis of pelvic bone tumors. *Bulletin of Electrical Engineering and Informatics*, 13(3), Article 3. <https://doi.org/10.11591/eei.v13i3.6861>
- Simonyan, K., & Zisserman, A. (2015). *Very Deep Convolutional Networks for Large-Scale Image Recognition* (No. arXiv:1409.1556). arXiv. <http://arxiv.org/abs/1409.1556>
- Suganyadevi, S., Seethalakshmi, V., & Balasamy, K. (2021). A Review on Deep Learning in Medical Image Analysis. *International Journal of Multimedia Information Retrieval*, 11(1), 19–38. <https://doi.org/10.1007/s13735-021-00218-1>
- Szegedy, C., Ioffe, S., Vanhoucke, V., & Alemi, A. (2016). *Inception-v4, Inception-ResNet and the Impact of Residual Connections on Learning* (No. arXiv:1602.07261). arXiv. <http://arxiv.org/abs/1602.07261>
- Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., Erhan, D., Vanhoucke, V., & Rabinovich, A. (2014). *Going Deeper with Convolutions* (No. arXiv:1409.4842). arXiv. <http://arxiv.org/abs/1409.4842>
- Tan, M., & Le, Q. V. (2020). *EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks* (No. arXiv:1905.11946). arXiv. <https://doi.org/10.48550/arXiv.1905.11946>
- Tan, M., & Le, Q. V. (2021). *EfficientNetV2: Smaller Models and Faster Training* (No. arXiv:2104.00298). arXiv. <https://doi.org/10.48550/arXiv.2104.00298>
- Usuzaki, T., Takahashi, K., Inamori, R., Morishita, Y., Takagi, H., Shizukuishi, T., Toyama, Y., Abe, M., Ishikuro, M., Obara, T., Majima, K., & Takase, K. (2024). Grading diffuse glioma based on 2021 WHO grade using self-attention-base deep learning architecture: Variable Vision Transformer (vViT). *Biomedical Signal Processing and Control*, 91, 106001. <https://doi.org/10.1016/j.bspc.2024.106001>

- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is All you Need. *Advances in Neural Information Processing Systems*. <https://dl.acm.org/doi/pdf/10.5555/3295222.3295349>
- Wang, J., Lu, S.-Y., Wang, S.-H., & Zhang, Y.-D. (2024). RanMerFormer: Randomized vision transformer with token merging for brain tumor classification. *Neurocomputing*, 573, 127216. <https://doi.org/10.1016/j.neucom.2023.127216>
- Xu, Y., & Meng, Z. (2024). Interpretable vision transformer based on prototype parts for COVID-19 detection. *IET Image Processing*, 18(7), 1927–1937. <https://doi.org/10.1049/ipr2.13074>
- Yadav, S. S., & Jadhav, S. M. (2019). Deep Convolutional Neural Network Based Medical Image Classification for Disease Diagnosis. *Journal of Big Data*, 6(1). <https://doi.org/10.1186/s40537-019-0276-2>
- Yue, Y., & Li, Z. (2024). *MedMamba: Vision Mamba for Medical Image Classification* (No. arXiv:2403.03849). arXiv. <http://arxiv.org/abs/2403.03849>
- Zhao, W., Chen, W., Li, G., Lei, D., Yang, J., Chen, Y., Jiang, Y., Wu, J., Ni, B., Sun, Y., Wang, S., Sun, Y., Li, M., & Liu, J. (2024). GMILT: A Novel Transformer Network That Can Noninvasively Predict EGFR Mutation Status. *IEEE Transactions on Neural Networks and Learning Systems*, 35(6), 7324–7338. *IEEE Transactions on Neural Networks and Learning Systems*. <https://doi.org/10.1109/TNNLS.2022.3190671>
- Zheng, S., Lu, J., Zhao, H., Zhu, X., Luo, Z., Wang, Y., Fu, Y., Feng, J., Xiang, T., Torr, P. H. S., & Zhang, L. (2021). Rethinking Semantic Segmentation from a Sequence-to-Sequence Perspective with Transformers. *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 6877–6886. <https://doi.org/10.1109/CVPR46437.2021.00681>
- Zhu, X., Su, W., Lu, L., Li, B., Wang, X., & Dai, J. (2021). *Deformable DETR: Deformable Transformers for End-to-End Object Detection* (No. arXiv:2010.04159). arXiv. <http://arxiv.org/abs/2010.04159>
- Zoph, B., & Le, Q. V. (2017). *Neural Architecture Search with Reinforcement Learning* (No. arXiv:1611.01578). arXiv. <https://doi.org/10.48550/arXiv.1611.01578>